

Week 12 — 🎨 Turtle Art

Name: _____ Date: _____

*Spirograph recipe: **draw a shape, turn a little, repeat** — it closes perfectly when the turns total **360°**. And `colors(mod(i, 6) + 1)` makes six pens paint forever.*

Today's words

Word	What it means
<code>pencolor()</code>	the pen's ink: 'r' 'g' 'b' 'c' 'm' 'y' 'k' — or mix [1 0.5 0]
<code>turtle_speed(0)</code>	zero pause per step — the turtle teleports
spiral	<code>forward(i)</code> — every side one step longer than the last
spirograph	a shape rotated again and again until the turns fill 360°
<code>colors(mod(i, 6) + 1)</code>	mod wraps the count so the six-pen string cycles forever

1 • Match the code to the art 🎨

Write the letter of each snippet next to its picture (each is used once).

- | | |
|---|--|
| A) <pre>for i = 1:4 forward(100) right(90) end</pre> | B) <pre>for i = 1:100 forward(i) right(90) end</pre> |
| C) <pre>for i = 1:5 forward(100) right(144) end</pre> | D) <pre>for i = 1:36 (draw a square) right(10) end</pre> |

- ____ a five-pointed star
- ____ one plain square
- ____ squares spiraling outward, each side longer than the last
- ____ a round flower/mandala made of 36 tilted squares

2 • The rotation table 🔄

Repeats to close = **360 ÷ tilt**. Fill in the blanks.

tilt after each shape	repeats to close
10°	36
20°	_____
30°	_____
45°	_____
120°	_____

3 • Design your own spirograph 🖍️ (we run these NEXT WEEK!)

- Shape sides (3–8): ____ • Tilt angle: ____° • **Repeats to close:** ____
- Colors, in order (pick up to 6 letters from r g b c m y k): ____
- Sketch your prediction:

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+-----+
|       |
|       |
|       |
+-----+
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🧠 Brain teaser (optional — take it home)

Neat spirographs need tilt angles that **divide 360 evenly**. 1, 2, 3, 4, 5, and 6 all work... 7 doesn't! How many divisors of 360 can you find? (There are **24**.) Bonus pondering: ancient astronomers *chose* 360 for the circle — why was that such a convenient number to pick?